

SANKAR RAGHUTHAMAN

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EDUCATION

North Carolina State University Master of Computer Science Raleigh, NC	Aug 2024 – May 2026
Mumbai University Bachelor of Engineering, Information Technology Mumbai, India	Aug 2016 – May 2020

PROFESSIONAL EXPERIENCE

NC State University Graduate Research Intern Raleigh, NC	Sept 2025 – Mar 2026
- Engineered two-phase RAG pipeline for code documentation combining intent-matched retrieval from open-source corpus with call graph context from same codebase, validated through controlled experiment on 7,976 Ruby methods across 7 repositories. - Curated 7,976 code-comment pairs by quality and intent using LLaMA-3-8B; validated against human annotations achieving Kappa 0.81, confirming label reliability and enabling high-confidence retrieval pools. - Validated pipeline through controlled human studies: intent-matched retrieval preferred 92.2% of time over unfiltered retrieval; combined pipeline improved comment quality 90% across readability, accuracy, and grounding. Accepted at Frontiers in Education 2026.	

Children's Hospital of Philadelphia Data Science Intern Philadelphia, PA	Jun 2025 – Dec 2025
- Revamped genomic classification pipeline for Genetic Syndrome (~200 patients, 850K+ methylation features), eliminating data leakage and replacing ill-suited ML approach, restoring clinical trust in results. - Constructed nested cross-validation (5-fold outer, 3-fold inner) with KNN imputation, BH feature selection reducing 850K features to ~200, and PCA scoped strictly inside each fold, preserving class-specific signal and producing unbiased estimates clinical team could act on. - Trained LinearSVC achieving 98-99% precision and 100% recall; adopted as lab standard by clinical researchers at CHOP.	

Tata Consultancy Services Mumbai, India	Aug 2020 – Jul 2024
<i>Machine Learning Engineer II</i> - Developed async LLM pipeline for extracting structured fields from 1,000+ weekly invoices; validated extractions using edit distance against pre-labeled data and embedding similarity to catch field assignment errors, achieving 95% precision. - Replaced brittle rule-based template system with confidence-based routing pipeline, cutting processing time 10x (50 to 5 minutes) and human review from 100% to 5% across multi-column and multilingual invoice formats. - Architected reference MLOps framework across 40+ pipelines on Azure Databricks with ML-specific CI/CD gates (schema validation, quality thresholds, drift detection), reducing pipeline failures by 80% and deployment cycles by 40%. - Implemented automated drift detection via Azure Monitor and MLflow versioned model registry, reducing issue detection from hours to under 5 minutes and enabling fully auditable deployments across all environments.	

<i>Machine Learning Engineer</i> - Deployed real-time FX anomaly detection using Isolation Forest on Spark Structured Streaming, replacing manual transaction review with continuous unsupervised scoring across multiple currency pairs without requiring labeled fraud data. - Diagnosed and resolved production partition skew (USD/EUR at 80% volume) via composite key repartitioning, increasing throughput 25% and eliminating stale-data incidents. - Introduced MLflow experiment tracking and versioned model registry as team's first standardized release process, establishing repeatable deployment standards later scaled across 40+ pipelines in the MLOps migration.	
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SKILLS SUMMARY

LLM and Context Engineering: RAG, GraphRAG, Multi-Agent Orchestration, Context Pipelines, Context Window Optimization, Tokenization, Few-shot Prompting, Chain-of-Thought, Conversational Memory, LLM-as-judge, Prompt Engineering, LangChain, LangGraph, ChromaDB, FAISS

Languages and Libraries: Python, SQL, Pandas, NumPy, PyTorch, scikit-learn, TensorFlow, Hugging Face Transformers, BERT, Sentence Transformers, asyncio, OpenAI API, Git

MLOps and Cloud: MLflow, Docker, Kubernetes, Azure Databricks, Azure Data Factory, Azure DevOps, Azure Monitor, GitHub Actions, CI/CD, Model Registry, Drift Detection

Data and Compute: PySpark, Spark Structured Streaming, Flask, REST APIs, Tree-sitter AST, Batch and Real-time Inference

TECHNICAL PROJECTS

Conversation Factory [GitHub]
- Built multi-agent pipeline (Scenario Planner, Assistant, User Agent, Mock Executor) generating synthetic multi-turn tool-calling conversations for LLM fine-tuning at scale. - Designed tool graph enforcing valid multi-tool API call sequences before generation, eliminating invalid chains that corrupt model training. - Applied LLM-as-judge scoring with repair loop and diversity steering, improving multi-step ratio from 59% to 64% with no quality regression.

Contract Analyzer
- Architected LangGraph platform routing 3 parallel agents (Clause, Risk, Summarizer) to automate contract analysis, cutting review time ~3x. - Implemented multi-query RAG with citation markers linking LLM claims to exact PDF passages, making contract insights auditable and traceable. - Added bilingual drift agent and configurable compliance scoring against client rulesets, enabling enterprise onboarding without code changes.

RESEARCH PUBLICATIONS

- "Classification of Disease using CBC," Springer Artificial Intelligence and Soft Computing. Jul 2021
- "WIP: Usefulness-Gated, Intent-Aware Code Comment Generation with Large Language Models," Frontiers in Education 2026. [Accepted]